

SUMS OF POWERS VIA CENTRAL FINITE DIFFERENCES AND NEWTON'S FORMULA

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ABSTRACT. In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the central Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

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1. AUXILIARY LEMMAS

In this section, allow us to define a few lemmas, definitions, and conventions that serve as foundation for main results of this manuscript. For multifold sums of powers, we use the notation provided by Donald Knuth in [?], because it is simple and beautiful

Definition 1.1 (Multifold sums of powers). *For non-negative integers r, n, m*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

We utilize the following notation for falling factorials

Definition 1.2 (Falling factorials). *For integers x, n*

$$(n)_k = \begin{cases} 1, & \text{if } k = 0, \\ n(n-1)(n-2) \cdots (n-k+1) = \prod_{j=0}^{k-1} (n-j), & \text{if } k > 0, \\ 0, & \text{else.} \end{cases}$$

We utilize the following notation for central factorials

Definition 1.3 (Central factorials). *For integers n, k*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n \left(n + \frac{k}{2} - 1\right) \left(n + \frac{k}{2} - 2\right) \cdots \left(n - \frac{k}{2} + 1\right) = n \left(n + \frac{k}{2} - 1\right)_{k-1}, & \text{if } k > 0, \end{cases}$$

where $\left(n + \frac{k}{2} - 1\right)_{k-1} = \prod_{j=0}^{k-2} \left(n + \frac{k}{2} - 1 - j\right)$ are falling factorials.

We encourage the audience to read J. F. Steffensen's work on the definition of generalized factorials [?].

Lemma 1.4 (Central Newton's formula). *For a function f defined on integers and arbitrary integers x, t*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t),$$

where $\delta^k f(t)$ denotes the k -th central finite difference

$$\delta^k f(t) = \sum_{j=0}^k (-1)^j \binom{k}{j} f\left(t + \frac{k}{2} - j\right),$$

and $(x-t)^{[k]}$ are central factorials defined by (1.3).

Proof. See [?, p. 462], and [?, section 2, eq. (19)], and [?, section 6, eq. (21)]. □

We have

Lemma 1.5 (Newton's formula for powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$n^m = \sum_{k=0}^m \frac{(n-t)^{[k]}}{k!} \delta^k t^m.$$

Proof. Special case of Lemma (1.4) for $f(n) = n^m$. □

We have

Lemma 1.6 (Central factorials binomial form). *For integers n and $k \neq 0$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Proof. We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!} \binom{n + \frac{k}{2} - 1}{k-1}_{k-1} = \frac{n}{k(k-1)!} \binom{n + \frac{k}{2} - 1}{k-1}_{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1},$$

because of the identity in falling factorials $\frac{(x)_n}{n!} = \binom{x}{n}$. □

We have

Lemma 1.7 (Generalized hockey-stick identity). *For arbitrary integers a, b and j ,*

$$\sum_{k=a}^b \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^b \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right)$. By hockey-stick identity $\sum_{k=0}^n \binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^b \binom{k}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. $\square$

We have

Lemma 1.8 (Centered hockey-stick identity). *For integer $n \geq 1$, and arbitrary integers t, k, r*

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Proof. By setting $a = 1 - t + \frac{k}{2} + r$, and $b = n - t + \frac{k}{2} + r$ into Lemma (1.7) yields

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

This completes the proof. $\square$

We have

Lemma 1.9 (Multifold sums of zero powers). *For non-negative integers r, n*

$$\Sigma^r n^0 = \binom{r+n-1}{r}.$$

Proof.

- (1) Let $r = 0$, then we have $\Sigma^0 n^0 = 1$, by definition (1.1).
- (2) Let $r = 1$, then we have $\Sigma^1 n^0 = \sum_{k=1}^n 1 = \binom{n}{1}$.
- (3) Let $r = 2$, then we have $\Sigma^2 n^0 = \sum_{k=1}^n \binom{k}{1} = \binom{n+1}{2}$.
- (4) Let $r = 3$, then we have $\Sigma^3 n^0 = \sum_{k=1}^n \binom{k+1}{2} = \binom{n+2}{3}$.

(5) Similarly, for arbitrary integer r , by hockey-stick identity $\sum_{k=1}^n \binom{k}{r} = \binom{n+1}{r+1}$, yields

$$\Sigma^r n^0 = \sum_{k=1}^n \Sigma^{r-1} k^0 = \sum_{k=1}^n \binom{k+r-2}{r-1} = \binom{r+n-1}{r}.$$

This completes the proof. □

We follow the convention

Convention 1.10. *For all x*

$$x^0 = 1.$$

Donald Knuth give extensive discussion on the convention $x^0 = 1$ for all x , including zero, in Concrete Mathematics [?, p. 162], and in [?].

2. INTRODUCTION

3. MAIN RESULTS

4. CONCLUSIONS

In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the central Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

MATHEMATICA PROGRAMS

Use this [GitHub](#) repository to validate the results using Mathematica programs.

Mathematica Function	Validates / Prints
<code>ValidateBackwardNewtonsFormulaForPowers.txt</code>	Validates Lem. (??)

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